

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA1558

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects.
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

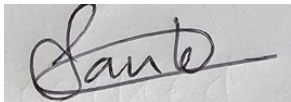
Total Marks: 30

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SANDEEP K / 192521253

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1. Website clicks, mobile interactions and transactions

The platform needs high-volume ingestion and separate handling of structured transactions and semi-structured interaction data. The architecture can be inferred from scale and reporting needs.

- Use distributed streaming plus batch/ETL ingestion.
- Store raw data in a scalable lake/HDFS layer.
- Use schemas/catalogs for structured and semi-structured data.
- Partition by date, region or business domain.
- Expose curated data to BI dashboards.

Reverse-Engineered Big Data Flow



2. E-commerce personalization

Real-time personalization requires synchronized customer, session, clickstream and transaction data. Fast profile access and scoring are likely priorities.

- Capture events through streaming.
- Use distributed customer/profile storage.
- Join sessions and orders with stable identifiers.
- Use caching and distributed query engines.
- Train models and perform real-time recommendation scoring.

Reverse-Engineered Big Data Flow



3. Healthcare histories, images, sensors and prescriptions

Healthcare combines structured records, large unstructured files and continuous streams. A hybrid architecture is therefore likely.

- Use relational storage for clinical records.
- Use distributed/object storage for images and files.
- Connect hospitals, labs and insurers through APIs/ETL.
- Use streaming plus ML for monitoring and prediction.
- Enforce encryption, access control, auditing and privacy.

Reverse-Engineered Big Data Flow

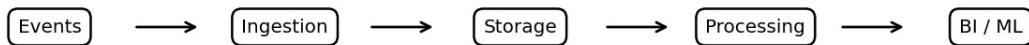


4. Smart city traffic and transport feeds

Continuous sensor feeds suggest edge preprocessing, distributed messaging and centralized historical storage. Geospatial and time-series processing support congestion analysis.

- Use edge nodes to filter and preprocess.
- Use distributed messaging for continuous feeds.
- Apply streaming analytics for congestion prediction.
- Use time-series and geospatial databases.
- Combine live dashboards with batch planning analytics.

Reverse-Engineered Big Data Flow



5. Healthcare analytics integration

The integrated platform must combine multiple source types while preserving governance. Distributed processing supports population-scale analysis and ML.

- Use hybrid storage for structured and unstructured data.
- Use ETL/interoperability pipelines and common IDs.
- Use streaming for wearable monitoring.
- Use batch processing for population studies.
- Apply encryption, audit, compliance and ML risk scoring.

Reverse-Engineered Big Data Flow

